

Network Intelligence Gathering Activity

Scenario:

You are a security analyst for a company that has recently added several new systems to its internal network. Before conducting any further security assessment, you need to identify which hosts are currently active on the subnet 192.168.10.0/24.

Task:

Perform a ping sweep on the network 192.168.10.0/24 using Kali Linux to discover all live hosts. Record the IP addresses of the active systems and prepare a brief report listing the reachable hosts. Based on your findings, explain how ping sweep results can assist an attacker during the reconnaissance phase and discuss two security measures that can reduce the effectiveness of ping sweeps on an organization's network.

Expected Outcome:

Identification of active hosts on the subnet.

List of reachable IP addresses.

Analysis of reconnaissance value.

Recommended defensive measures against host discovery.

Step 1: Verify Network Configuration

Open the Kali Linux terminal and verify that the system is connected to the target network.

Command:

```
ip addr
```

Step 2: Perform a Ping Sweep

Use Nmap to discover all live hosts on the network.

Command:

```
sudo nmap -sn 192.168.10.0/24
```

The -sn option performs host discovery only and does not scan ports.

Observe the Output

The scan identifies the devices that respond to ping requests.

Sample Output:

```
kali@kali: ~  
Session Actions Edit View Help  
[kali@kali]~  
$ ip addr  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def  
ault qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g  
roup default qlen 1000  
    link/ether 08:00:27:d8:ae:f2 brd ff:ff:ff:ff:ff:ff  
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0  
        valid_lft 86226sec preferred_lft 86226sec  
    inet6 fd17:625c:f037:2:6472:dc48:ef8a:1bcb/64 scope global temporary dyna  
mic  
        valid_lft 86228sec preferred_lft 14228sec  
    inet6 fd17:625c:f037:2:a00:27ff:fed8:aef2/64 scope global dynamic mngtmpa  
addr noprefixroute  
        valid_lft 86228sec preferred_lft 14228sec  
    inet6 fe80::a00:27ff:fed8:aef2/64 scope link noprefixroute  
        valid_lft forever preferred_lft forever  
[kali@kali]~  
$ sudo nmap -sn 192.168.10.0/24  
[sudo] password for kali:  
123Sorry, try again.  
[sudo] password for kali:  
Sorry, try again.  
[sudo] password for kali:  
sudo: 3 incorrect password attempts  
[kali@kali]~  
$ sudo nmap -sn 192.168.10.0/24  
[sudo] password for kali:  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-07-28 23:37 -0400  
Nmap scan report for 192.168.10.0  
Host is up (0.015s latency).  
Nmap scan report for 192.168.10.1  
Host is up (0.0033s latency).  
Nmap scan report for 192.168.10.2  
Host is up (0.010s latency).  
Nmap scan report for 192.168.10.3  
Host is up (0.010s latency).  
Nmap scan report for 192.168.10.4  
Host is up (0.0026s latency).  
Nmap scan report for 192.168.10.5  
Host is up (0.0031s latency).  
Nmap scan report for 192.168.10.6  
Host is up (0.0030s latency).  
Nmap scan report for 192.168.10.7
```

Step 5: Analyze the Results

The ping sweep successfully identified the active systems connected to the 192.168.10.0/24 subnet. These hosts are reachable and can be selected for further security assessment, such as port scanning, service enumeration, or vulnerability analysis.

Step 6: Reconnaissance Value

A ping sweep is commonly used during the reconnaissance phase to identify live systems on a network. It helps attackers or security analysts determine which hosts are active, reducing the number of systems that need further scanning. This information can be used for additional security testing or vulnerability assessment in an authorized environment.

Step 7: Defensive Measures

1. Restrict ICMP traffic using firewalls to prevent systems from responding to ping requests.
2. Use network segmentation and IDS/IPS to detect scanning activity and limit the spread of reconnaissance across the network.

Result

The ping sweep on the **192.168.10.0/24** subnet was completed successfully using Kali Linux. Active hosts were identified and recorded. The activity demonstrated the role of ping sweeps in network reconnaissance and highlighted defensive measures such as ICMP filtering, network segmentation, and IDS/IPS to reduce the effectiveness of host discovery.